

Exploring Psychologist Applied Biomarkers in Bipolar Disorder: A Systematic Framework

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Abstract. Bipolar disorder (BD) is a multifaceted mental illness characterised by alternating manic or hypomanic and depressed episodes. The present diagnosis is predominantly based on subjective assessments, frequently resulting in delays and misdiagnosis because of symptom overlap with other conditions. Notwithstanding progress in biomarker research, their incorporation into clinical practice is still constrained, underscoring a significant deficiency in objective diagnostic instruments for BD. This study examined qualitative interviews with seven mental health professionals to determine essential biomarker criteria and propose enhancements for bipolar disorder diagnosis. Research indicates constraints in subjective diagnosis methods and underscores the promise of integrating digital tools with biomarkers to improve precision. Essential characteristics for data-driven decision-making were identified to facilitate early detection and action. The study advocates for user-friendly digital health tools for clinicians that incorporate biomarkers to enhance the diagnosis and management of bipolar disorder. These insights facilitate the development of a paradigm aimed at improving long-term patient outcomes.

Keywords: Bipolar disorder · Biomarkers · Digital health tools · Diagnostic accuracy · Early diagnosis · Patient engagement

1 Introduction

BD is characterized by alternating episodes of mania or hypomania and depression, severely impairing daily functioning [31]. It impacts almost 50 million individuals worldwide and is linked to considerable morbidity and death, including elevated risks of suicide and comorbidities such as diabetes and cardiovascular illnesses [27, 25]. Contemporary diagnostic techniques, based on DSM-5 and ICD criteria, are constrained by the absence of objective biomarkers, resulting in diagnostic delays and mistakes [13].

The present diagnosis of BD predominantly depends on subjective evaluations, potentially resulting in biases and delays [10, 16]. Despite the potential of biomarker research, its incorporation into clinical practice is constrained due to the absence of objective methods for precise and early diagnosis [18, 37, 1, 22]. This gap obstructs prompt business development identification. Existing

techniques are hampered by the paucity of objective biomarkers and holistic diagnostic frameworks [14, 29, 3]. This study analyses physicians’ views on using biological and behavioural biomarkers in diagnostic instruments to better early bipolar illness diagnosis and treatment, developing a more objective diagnostic framework. A qualitative review of mental health experts’ data suggests design changes to increase diagnostic accuracy [2].

This research involved semi-structured interviews with seven mental health specialists to ascertain essential biomarkers for the diagnosis of bipolar illness, encompassing HRV, inflammatory markers, oxidative stress markers, genetic markers, neuroimaging alterations, and lifestyle factors. Thematic analysis of the interviews yielded insights into the application of biomarkers in clinical practice and guided recommendations for a digital system that integrates various biomarkers to enhance diagnostic accuracy and patient involvement.

This work underscores the constraints of existing BD diagnostic techniques, which predominantly depend on subjective symptom evaluations. Through consultations with clinicians, we investigate the potential of merging biological and behavioural markers with digital tools to enhance the early detection and management of bipolar disorder. We identify critical indicators, including heart rate variability, inflammatory markers, oxidative stress markers, genetic markers, neuroimaging alterations, and patient engagement measures, and delineate fundamental characteristics for novel diagnostic instruments. These innovations seek to improve diagnostic accuracy, allowing clinicians to make more precise and quicker judgments, hence enhancing patient outcomes. In conclusion, we present the following contributions to this study:

1. We introduce a holistic framework integrating biological, behavioral, and psychosocial factors to enhance BD management through advanced technologies.
2. We explore how technology—like AI monitoring, wearables, and gamified apps can improve BD diagnosis and management.
3. Through clinician interviews, we analyze their expectations and preferences for BD diagnostic tools.
4. We identify essential biomarker measures and patient engagement data that can be incorporated into digital diagnostic tools for BD.
5. We highlight key design aspects for digital health tools to boost diagnostic accuracy, improve patient outcomes, and address BD differentiation challenges.

2 Related Work

The diagnosis of bipolar disorder (BD) has conventionally depended on subjective clinical evaluations according to DSM-5 and ICD criteria. Although these guidelines offer a framework, their reliance on clinician observations and patient self-reports frequently results in misdiagnosis, especially in differentiating bipolar disorder from unipolar depression [31, 5, 10]. Diagnostic delays may lead to

years of unsuitable treatment prior to the establishment of an accurate BD diagnosis [7, 9].

To mitigate these limitations, researchers have investigated biomarkers including heart rate variability (HRV), cortisol levels, and neuroimaging data, which provide objective insights into the physiological foundations of mood dysregulation in bipolar disorder (BD) [33, 25, 19]. Nonetheless, despite their promise, these biomarkers have experienced sluggish clinical adoption owing to difficulties in their integration into current diagnostic frameworks and the intricacy of amalgamating multiple indicators into a cohesive instrument [5, 16, 8, 35]. Recent advancements in digital health technologies, particularly wearable devices, facilitate continuous monitoring of behavioural and physiological indicators, including sleep patterns, physical activity, and social interactions, all pertinent to the management of bipolar disorder. Mobile health applications augment long-term monitoring and intervention by integrating real-time data with clinical evaluations [15]. Nonetheless, inadequate integration into clinical processes and insufficient clinician participation in tool development continue to pose substantial obstacles to widespread use [11, 28]. Research indicates that clinician-centered design, which integrates iterative feedback from physicians, enhances tool use and acceptance [20, 17, 12].

Machine learning (ML) methodologies, especially those employing neuroimaging and cognitive evaluations, have demonstrated significant accuracy in differentiating BD from healthy controls [6, 24]. Nonetheless, numerous ML models neglect to consider clinical variability, hence constraining their practical usefulness in real-world scenarios [23, 26]. Addressing this disparity necessitates diagnostic frameworks that integrate biomarkers, digital phenotyping, and clinical experience to guarantee both scientific rigour and practical applicability.

Our project tackles these problems by creating a clinician-focused diagnostic instrument that incorporates biological and behavioural indicators, facilitating smooth integration into clinical practice and enhancing diagnostic accuracy.

3 Method

This project aims to investigate the significance of biomarker measures. To examine the importance of biomarker assessments in diagnosing and treating BD, we performed semi-structured interviews with seven mental health professionals. The interviews sought to comprehend existing diagnostic methods, encountered problems, and anticipations for forthcoming diagnostic instruments. Experts specialising in the diagnosis and treatment of BD were requested to contemplate their experiences, pinpoint essential biomarkers, and propose enhancements for digital health systems. This qualitative methodology offered insights into the incorporation of biomarkers into diagnostic frameworks to improve accuracy and efficiency.

3.1 Interview Study

Clinicians Seven mental health professionals engaged in this study, concentrating on the diagnosis, treatment, and management of bipolar disorder (BD). These clinicians, who also conduct pertinent research, offered insights into critical biomarkers and the constraints of existing diagnostic techniques. Their expertise was essential in determining how biomarker parameters could enhance diagnostic precision and patient outcomes. The study sought to ascertain their expectations for forthcoming digital health technology to improve diagnostic tools. Our study included seven clinicians, a limited sample size for quantitative

Table 1. Clinicians’ Information.

ID	Specialization	Experience (years)	BD Patients/Week
C01	Psychiatry	6	5
C02	General Medicine	5	5
C03	Psychiatry	6	4
C04	Psychiatry	4	5
C05	Psychiatry	5	7
C06	Neurology	6	6
C07	General Medicine	3	4

research yet appropriate for qualitative research. We prioritised comprehensive insights over quantity, in accordance with Burawoy’s focus on deriving profound understanding from a restricted number of examples [34]. We concentrated on professionals possessing practical expertise in BD diagnosis and treatment, as well as a receptiveness to innovative technology. Data saturation was attained by uniform experiences and discoveries concerning biomarkers and diagnostic instruments.

Procedure Interviews were conducted from May 12 to August 27, 2024, using a semi-structured approach. Questions focused on:

1. Current Processes and Challenges: Steps and challenges in detecting and diagnosing BD.
2. Key Symptoms and Monitoring: Symptoms, lifestyle factors, and current monitoring techniques.
3. Technological Integration: Potential of AI and biomarker tools in clinical practice.
4. Advice for Researchers: Expert opinions on technology-based solutions and biomarker potential.

3.2 Data Analysis

Thematic analysis was performed utilising both inductive and deductive coding methodologies. The inductive phase facilitated the organic emergence of patterns

and themes from the data, directly recording physicians’ insights. The deductive phase subsequently employed existing ideas to systematically analyse these themes. This dual methodology integrated empirical data with theoretical classification, resulting in a thorough comprehension. Sub-themes were derived from distinct codes, which were subsequently synthesised into principal themes. [2].

Table 2. Main Themes in Research.

Main Theme	Description
Current Process and Challenges	It highlights the specific challenges they face in their day-to-day work.
Exploring Psychologist-Applied Biomarker Parameters	Using biomarkers in clinical practice for mental health assessment.
Prototype	Design and functionality suggestions for diagnostic prototypes

4 Results

4.1 Current Assessment Process of BD

A complete medical history, psychiatric symptoms, and family mental health history are needed to diagnose bipolar disorder. Clinicians can diagnose manic and depressive episodes by understanding their cyclical character. “Mood swings are crucial for identifying behavioural changes”, said a clinician. Evaluations track mood changes over weeks or months to document the disorder’s usual variations. “Identifying the pattern of mood episodes is crucial for distinguishing bipolar disorder from other conditions,” said another. Clinicians use DSM-5 criteria and diagnostic tools like the YMRS for manic symptoms and the HAM-D for depressive symptoms. They improve clinical observations with quantitative measurements. Structured interviews are needed to collect symptom frequency, duration, and intensity. “Structured interviews facilitate the collection of comprehensive data that may be overlooked in casual discussions,” stated a clinician. Obtaining meaningful patient information, especially during manic periods when self-awareness may be low, is difficult. Thus, clinicians commonly use family information to complete their clinical picture. “Patients frequently underreport their manic symptoms, rendering family collateral an essential element of the assessment,” said a clinician (C04). Because many symptoms resemble other mental illnesses, notably major depressive disorder, diagnosis is often uncertain. Different symptom presentations, episode strength and duration, comorbid disorders, and social conditions may hinder help-seeking. The symptom overlap between bipolar disorder and other mood disorders makes diagnosis complicated and time-consuming, noted a clinician (C01). Clinical considerations include sleep quality and consistency mood episode symptoms and factors. Diagnostic signs include social disengagement and activity alterations. A coach noted sleep

issues often signal a mood disorder. Continuous management monitors therapeutic efficacy and mood episode recurrence. Later assessments include medication efficacy, side effects, emotional symptom stability, functional recovery, and relapse symptoms. Changing treatment regimens and preventing relapses require regular follow-ups, noted one clinician (C07).

4.2 Parameters Used

Biological and Behavioral Markers: Clinicians acknowledge physiological and behavioural indicators as crucial to understanding and treating BD. Clinicians use HRV and EDA to predict stress and mood swings in bipolar disorder (BD). “HRV is a key indicator of autonomic nervous system dysfunction and mood instability,” (C01). HRV shows autonomic nervous system function and aids mood assessment. To reduce false positives in AI-based risk classification, clinicians emphasise its importance in combination with sleep and exercise. “Declines in HRV frequently precede manic episodes; however, contextual factors—such as sleep disturbances—are crucial to validate the trend,” noted another clinician (C03). Electrodermal activity patterns are crucial as they signify fluctuations in emotional arousal and can enable early intervention. “Electrodermal reactivity may signify an impending manic episode, facilitating prompt intervention,” asserted a clinician (C04). Besides physiological markers, lifestyle factors, especially sleep patterns, are also examined. One clinician said, “Sleep disturbances often serve as the initial indicator of a mood episode and are essential in mood regulation.” Clinicians assess sleep quality and consistency as mood episode symptoms and causes. Wearables and smart mattresses with advanced sensors can monitor sleep. Physical activity and nutrition change significantly. “Variations in activity levels and appetite frequently precede mood alterations and serve as critical indicators for early intervention,” said another clinician (C05). Clinicians say hunger and activity changes can indicate mood disorders. Key diagnostic signs are activity changes and social disengagement. “Social withdrawal is a significant predictor of an approaching depressive episode, whereas heightened social engagement may indicate mania,” said a clinician (C06). Wearable and digital health instruments provide crucial factors for establishing effective BD monitoring and intervention strategies.

Wearable APIs allow real-time physiological data integration into clinical dashboards or AI systems. Clinicians stressed the need for EHR integration and protected cloud pipelines to protect data. Comfort, especially during continuous use, was emphasised, with lighter, less noticeable gadgets receiving more praise. To reduce patient burden and improve adherence, passive sensing and automated synchronisation devices are preferred.

Psychosocial Factors and Relationships: Effective business development management requires psychosocial and interpersonal considerations. Clinicians say family and close social networks are essential for mood stability and symptom management. Stabilising mood and managing symptoms require social support, especially from family and close friends. Isolation can worsen a patient’s condition, yet strong social interactions improve outcomes. Positive social contacts

Table 3. Biological, Behavioral, and Psychosocial Factors in Mental Health Assessment

Category	Subcategory	Description
Biological and Behavioral	Physiological Markers	HRV, EDA
	Emotional Symptoms	Dysregulation, mood fluctuations
	Cognitive Changes	Memory issues, impulsivity, risky behaviors
	Physical Symptoms	Sleep/appetite disturbances, activity changes
	Risk Behaviors	Self-harm, suicidality
	Lifestyle Factors	Nutrition, physical activity
Psychosocial	Social Support	Positive interactions, family involvement
	Social Challenges	Isolation, negative interactions
	Behavioral Patterns	Social withdrawal/over-engagement
	Therapeutic Support	Therapy, support groups
	Family History	Genetic predisposition, psychiatric history
	Caregiver Role	Monitoring mood changes

help regulate mood, says a clinician (C01). In contrast, negative social contact and seclusion can worsen symptoms, especially during depression. Depression is exacerbated by negative social interactions and isolation. Adverse social encounters might cause mood swings. (C02). Changes in social behaviour and engagement indicate mood swings. Increased social involvement often precedes manic episodes, while social retreat usually precedes depression. Depression is often accompanied by social disengagement (C03). Depression often leads to suicide thoughts and self-harm. Suicidal thoughts is widespread during depressive episodes, requiring close monitoring and management (C03). Therapeutic interactions and ongoing therapy provide mood regulation tools. “Therapeutic relationships provide consistent support and direction” (C02). Family and caretakers must monitor mood swings and provide emotional support. “Family members observe the patient’s mood and behavior” (C02). Finally, familial and medical history are needed to diagnose bipolar disorder, which may be inherited. Understanding familial mental health provides valuable insights (C05, C03). Therapeutic therapies for bipolar disorder must include psychosocial factors.

4.3 Harnessing Technology for BD Management: Ethical, Clinical, and Research Guidance from Clinicians:

This section outlines the findings from our discussions with clinicians regarding technology-based strategies for BD management. Three primary conclusions emerged: notable biomarkers and metrics for monitoring BD, ethical implications of technology, and suggestions for researchers. We propose a framework entitled

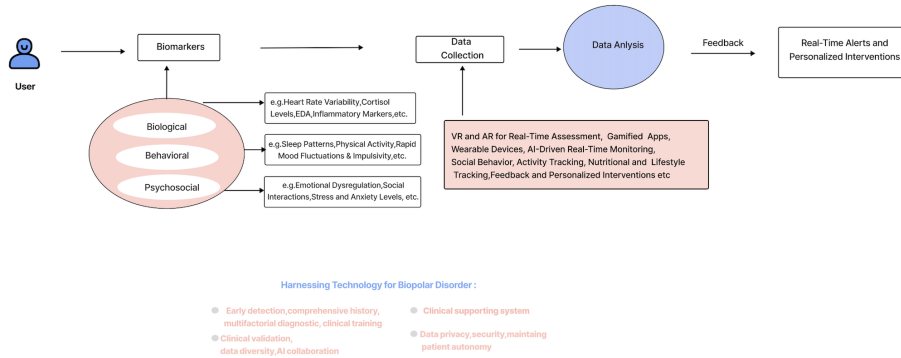


Fig. 1. Framework for managing BD using integrated biomarkers.

“Harnessing Technology for BD Management” to integrate these findings, highlighting the collection and analysis of biological, behavioural, and psychological data to provide personalised treatments while ensuring clinical validation, data protection, and compliance with ethical standards. Clinicians expressed substantial ethical apprehensions over the utilisation of AI and technology in the monitoring of bipolar disorder. They underscored the significance of patient permission, data security, and privacy, highlighting that patients must comprehend the methods of data collection, storage, and use. A clinician asserted, “Patients must comprehend the utilisation of their data and possess control over it” (C02). They emphasised the peril of algorithmic bias, observing that AI models trained on unrepresentative datasets may intensify health inequities. “It is imperative to validate AI models across diverse demographics to ensure equity,” asserted another clinician (C05). Concerns regarding the prolonged exploitation of data, including unauthorised access by third parties, were expressed. Clinicians advocated for comprehensive data governance frameworks featuring transparent consent processes and patient data control to resolve these challenges. They proposed employing federated learning and edge AI to improve privacy and ensure compliance with legislation like as GDPR and HIPAA. “Incorporating ethics into system design from the outset is essential for patient-centered and socially responsible care,” stated a physician (C07).

Essential Metrics for Monitoring Bipolar Disorder and Framework:

Clinicians stressed bipolar disorder monitoring requirements. HRV and GSR are crucial mood indicators. HRV and GSR can predict mood episodes, according to a clinician (C01). Wearable gear enhances quick detection. Sleep abnormalities like hypersomnia in depression and sleeplessness in mania are important markers. A clinician (C04) said, “Monitoring sleep patterns is crucial for predicting mood fluctuations.” Wearables and smart mattresses with advanced sensors track sleep. Changes in diet and exercise are significant. Clinicians recognised that appetite and activity changes can indicate mood episodes (C05, C06). A

neuro-behavioral feedback apparatus uses real-time neuroimaging and behavioral monitoring to deliver comprehensive insights. Emotional instability and impulsivity are equally important. Passive digital surveillance, including smartphone use and physical activity, helps identify behavioural anomalies. Treatment must address melatonin and circadian rhythm disorders. Melatonin and circadian rhythm monitoring are needed to stabilise bipolar disorder mood. Wearables and smart mattresses can track sleep and melatonin. Cognitive tests and financial monitoring can spot risky behaviour. Early mood episode detection requires monitoring oxidative stress and inflammatory indicators. Social behaviour can reveal mood swings early. Hunger, weight, and diet affect mood. Holistic care treats anxiety and substance misuse. Entertainment computing enhances diagnosis.

Guidance for Researchers: To guarantee the efficacy of technology-driven solutions for BD monitoring and therapy, researchers must prioritise clinical validation. A physician asserted, “Prototypes necessitate comprehensive clinical validation.” (C02). The multivariate data collection approach enhances precision. We propose a hybrid AI architecture that integrates long short-term memory (LSTM) networks for analysing sequential physiological signals (e.g., HRV, sleep, electrodermal activity) with dense feedforward layers for structured behavioural and self-reported data (e.g., social activity, mood logs). Data from several modalities are amalgamated using a late fusion technique employing a multi-head attention mechanism, allowing the model to assess and prioritise inputs according to their predictive significance for mood state transitions. Real-time inference is enabled by APIs integrated into wearable devices that transmit encrypted data to a cloud-based processing layer, which then provides risk scores and trend summaries to clinicians through a dashboard. Artificial intelligence should assist clinicians in decision-making without supplanting human competence. Usability, accessibility, and patient-centred design are essential for effective therapeutic interventions. Moreover, prioritising adaptability is essential to facilitate modifications informed by clinical feedback. A thorough symptom analysis across several data points substantiates any technology-driven intervention for BD.

5 Discussion

This study investigated the difficulties clinicians encounter in identifying bipolar disorder and how the incorporation of biomarker characteristics with computerised diagnostic tools can improve accuracy and treatment strategies.

Biomarker Integration for Diagnosis: Our research indicates that integrating essential biomarker traits into bipolar disorder diagnostic algorithms markedly enhances diagnostic precision and facilitates more individualised treatment. Clinicians recognised physiological indicators such as HRV, cortisol concentrations, and sleep patterns as essential data points, in addition to behavioural markers. This integration facilitates data-driven, real-time therapeutic decisions. These findings correspond with increasing studies endorsing the utilisation of objective physiological data in mental health diagnosis, namely heart rate variability.

ity as an indicator of mood instability [21, 32]. Biomarkers address the shortcomings of conventional techniques reliant on patient self-reports and professional observations. According to Perochon S (2023) and Boukarras S (2024), clinicians can identify mood fluctuations earlier and execute prompt therapies by observing physiological indicators in conjunction with behavioural alterations [4, 30]. Behavioural indicators, including alterations in physical activity and sleep patterns, serve as critical early warning signs of mood instability. Observing sleep patterns yields insights on the emergence of manic and depressive episodes. The integration of behavioural and biological markers assists clinicians in addressing the unpredictability of bipolar disorder symptoms. In conclusion, the amalgamation of behavioural and biological markers into a unified diagnostic framework offers a more methodical and objective method for detecting bipolar disease. This allows clinicians to expedite and customise treatment decisions through a data-driven approach [38, 36].

Current Diagnostic Challenges: Clinicians continue to have difficulties in diagnosing bipolar disorder, especially in distinguishing it from other mood disorders such as unipolar depression. The symptom overlap frequently results in misdiagnosis, with patients exhibiting analogous depressive symptoms in both contexts. Manic or hypomanic episodes, critical characteristics of bipolar disorder, are frequently concealed or underreported by patients. This may hinder physicians' ability to identify the complete range of BD symptoms. The patient's condition may deteriorate, and long-term care may become increasingly challenging due to this misdiagnosis, perhaps delaying necessary treatment for years. BD is challenging to diagnose due to the extensive variability in symptom presentation. Marked variations in mood, activity, and behaviour are characteristic of bipolar disorder. Some individuals may see intense, rapid mood fluctuations, whereas others endure significant periods between episodes. BD is frequently misinterpreted as unipolar depression or other mood disorders due to its initial presentation of milder or atypical symptoms. Further impediments to precise diagnosis encompass cultural stigma and patient hesitance to reveal symptoms. This hesitation may hinder clinicians from obtaining a comprehensive understanding of the patient's mood fluctuations, resulting in erroneous or inadequate evaluations.

Understanding Clinicians' Anticipations for Diagnostic Instruments: Clinicians identified essential criteria for diagnostic tools utilised in the treatment of BD, highlighting the need for systems that enable real-time monitoring, provide longitudinal data, and can be customised to meet individual patient requirements. Given the variable nature of bipolar disorder, physicians highlighted the importance of sleep patterns and HRV as essential physiological indicators for tracking sudden mood changes. The integration of wearable technologies was emphasised as a primary expectation. Wearable sensors provide the advantage of continuous, remote monitoring of physiological data, including heart rate variability, physical activity, and sleep patterns. The capacity to remotely gather data facilitates prompt modifications to medication or therapy according to current patient metrics, hence improving treatment efficacy. The advancement of

AI-driven diagnostic tools raises significant ethical concerns, especially with patient autonomy and data privacy. Clarity concerning the collection, retention, and utilisation of patient data is crucial. Clinicians articulated a significant expectation for the personalisation of diagnostic instruments. Given that each patient with BD exhibits distinct characteristics, tools must be tailored to address the individual requirements and situations of each person. Clinicians anticipate the integration of advanced technical components, such as wearable devices and real-time monitoring, with ethical measures to uphold patient autonomy and privacy in bipolar disorder diagnostic tools.

Designing a Clinician-Centered Diagnostic Prototype: The outcomes of this study emphasise that an effective clinician-centered diagnostic prototype for BD must integrate behavioural and biological indicators, facilitate real-time data processing, and provide an intuitive user interface. The physicians participating in the study underscored that the development of diagnostic tools should augment their clinical practices rather than interfere with them. The amalgamation of wearable technologies with AI platforms to assess both objective and subjective data is a crucial component of the proposed prototype. The technology allows clinicians to remotely and simultaneously monitor patients by continuously collecting real-time physiological data. The diagnostic prototype’s effectiveness would improve by integrating decision-support tools that provide actionable insights based on real-time data trends. Ensuring the interface is intuitive and user-friendly for physicians is a critical component of the prototype’s design. Furthermore, a crucial characteristic that practitioners desire in diagnostic tools is customisation. The prototype must offer individualised monitoring tailored to each patient’s specific needs. In summary, an effective clinician-centered diagnostic prototype should incorporate wearable technology for continuous data collection, real-time monitoring of behavioural and physiological variables, and decision-support tools that provide actionable insights while preserving physician autonomy.

6 Limitations and Future Directions

This study possesses some shortcomings that warrant acknowledgement. The limited sample size (N=7 clinicians) constrains the generalizability of the findings, and the absence of patient data from varied demographics raises concerns regarding broader application. Secondly, although the proposed biomarker integration system demonstrates potential, it necessitates validation in actual clinical environments. Third, significant ethical concerns around data protection and AI interpretability were not thoroughly examined. Future research should: (1) encompass larger, more heterogeneous samples comprising both clinicians and patients across various healthcare environments; (2) implement clinical trials to substantiate the efficacy of the diagnostic framework; and (3) examine essential ethical considerations, including data protection measures and patient consent procedures. Moreover, tailoring these tools to diverse cultural and socioeconomic situations will be crucial for equitable execution.

7 Conclusion

Bipolar disorder (BD) diagnosis using biomarker assessments and clinicians' hopes for future tools are examined in this study. Specialist interviews revealed physiological and behavioural signs that could improve diagnosis. The findings demonstrate the need for data-driven, real-time monitoring solutions that connect with clinical procedures and provide personalised insights.

Use an integrated biomarker-clinical framework, wearable tech for continuous monitoring, and AI-driven decision support to overcome diagnostic limits. We found that clinician-centered technologies increase BD detection precision and timing, enabling early intervention. These findings can improve biomarker efficacy in minimising misdiagnosis and improving patient outcomes in clinician-specific digital health systems. For successful acceptance, future research should focus on clinical validation, data privacy, and practical implementation. Using objective biomarkers and clinical evaluations could improve BD diagnosis, therapy, and long-term care.

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